Doc. No.: TD-LO40-DC-2002	Doc. Title: 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage And Generic Calibre DRC	Doc.Rev: 10	Tech Dev. Rev: 1.12.1_0

Document Level: (For Engineering & Quality Document/工程暨品质文件专用)

☐ Level 1 - Manual ☑ Level 2 - Procedure/SPEC/Report ☐ Level 3 - Operation Instruction

Security Level: ☐Security A - Top Secret ☑ Security B - SMIC Confidential

☐ Security C - SMIC Restricted ☐ Security D - SMIC Internal

Document Change History

Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
6	1.10_0	2015-09-08	Penny Hua	1. Update Title to 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low leakage and Generic Design Rule Check (DRC) runset of Calibre DRC; 2. Update Purpose description in section 2; 3. Update Reference in section 5; 4. Update responsibility in section 6; 5. Delete DRC tool versions in section 7, for the related content was added into item 8 6. Add DRC tool versions in section 8; 7. Add section 9 “Calibre DRC options set” to explain how to set the switch in DRC runset to customer; 8. Add item 10 for known issues; 9. Update attachment main rules' DRC runset to align with DR, as below: 1) Add switch #DEFINE MOM_5_6_CHECK, #DEFINE MOM_4_CHECK 2) Add rules MOM_4_G,MOM_5_G,MOM_6_G 3) Update below density rules: GT_31_density，M1_15_density，M2_14_density，M3_14_density，M3_16_M1M2M3_density，M4_14_density，M4_16_M2M3M4_density，M5_14_density，M5_16_M3M4M5_density，M6_14_density，M6_16_M4M5M6_density，M7_14_density，M7_16_M5M6M7_density，M8_14_density，M8_16_M6M7M8_density，IND_R_9_G_l_TM2_density,IND_R_9_G_h_TM2_densit, IND_R_9_G_l_MTT2_density, IND_R_9_G_h_MTT2_density 4) Update DNW_8 ALPA_1 LU rules,MTT2_13, BD_6,GT_20b,AA_C_5, Mn_12,Mn_13,GT_33 downgrade to [R] 5) Add DUMC_34a,DUMC_35a,SAB_16_R,NW_1a, TG_16a,GRC_7~GRC_12,LD_10,LD_11 6) Delete ULVT_3,ULVT_5 7) Update DUMC_34,DUMC_35,Mn_2,SRDOP_15,

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				NW_3,ULVT_4,ULVT_6,SAB_5,SAB_6,SAB_8,SAB_11, TG_16,GT_5a,TG_11_R,DUMC_26,CT_1,CT_8, CT_13, DNW_8 8) Move DG_15~DG_19 to TG_16~TG_20 9) Update AA_19,GT_20b,SAB_5,Convention_2,DG_16, DG_3,DG_4,TG_3,TG_4,GT_28,GT_32, GT_33, CT_21,SP_8c,SN_7c,SN_8c 10) Delete AA_9b,M1_4a,Mn_4a 11) Update AA_20,DG_18,DG_19,SP_7c,SN_7c, LU_3_3a/b/c/d,AA_15a_R, AA_15b_R, GT_20b, IND_R_8,GT_31, AA_9b, AA_13a, DIFRES_6, DNW_9,NW_11_R,NW_12_R,NW_17_R,RESNWST_6, RESNWAA_9,RESNWAA_11,DG_16,DG_17,DG_18,DG_19,GT_3e,GT_5,GT_5a,SRGT_5,GT_9,SRGT_10,GT_21,GT_28,GT_32,GT_33,PLRES_6,LVT_4,LVT_4a,LVT_5,LVT_5a,HVT_4,HVT_4a,HVT_5,HVT_5a,SP_6,SP_7a/b/c,SN_6,SN_7a/b,CT_5,CT_5_R,CT_20,M1_4a,Mn_4a,V1_19,Vn_19,Vn_23_R,PA_11,ESD_5,MTT2_11,SRAM device rule,LOGO_1,LU_3 12) Delete LVT_15,HVT_15,ULVT_10,SP_22,SN_23, SAB_8a/b, 13) Update switch follow 4R Design rule. 14) Add DFM priority for recommended rules. 15) Update connects follow net definition. 16) Implement ESD and Latch up rules 17) UpdateAA_1,AA_4a,AA_4b_4c,AA_9a,AA_11,AA_13,AA_14,AA_14a,AA_15a,AA_15b,AA_16,AA_17, AA_18,AA_19,RESAA_7,RESAA_12,RESAA_13, AA_C_1,AA_C_4,AA_C_5,DIFRES_6,DIFRES_7, DIFRES_11,DIFRES_12,DNW_1,DNW_2,DNW_2a, DNW_7,NW_1,SRNW_1,NW_3,NW_7,NW_11_R, NW_12_R,SRNW_14,NW_13,NW_14,NW_15, NW_16,RESNWST_6,RESNWST_7,RESNWST_9, RESNWAA_7,RESNWAA_9,RESNWAA_10,RESNWAA_11,PSUB_1,PSUB_4,PSUB_12,TG_1,TG_2,TG_13,TG_14,TG_15,DG_1,DG_2,DG_13,DG_14,BORDER_1,GT_2a,GT_3,GT_3c,SRGT_3,GT_4,SRGT_5,SRGT_6,SRGT_7,GT_8a,GT_17,GT_18,GT_20b,GT_21,GT_27,GT_28,RESP1_7,PLRES_6,PLRES_7,PLRES_8,PLRES_9,PLRES_10,LVT_1,LVT_13,HVT_1,HVT_13,SP_9,SRSP_10,SP_15,SP_20a,SN_7b,SN_9,SN_15,SAB_5,SAB_6,SAB_8,SAB_11,CT_1,CT_3,CT_3a,CT_8,CT_10,M1_1,M1_2,M1_3,SRM1_3sa,M1_4,M1_5,M1_6/6a/b/c,M1_7a,M1_7bc,M1_7

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				e,M1_8/8a,M1_9,M1_10,M1_14,V1_2,V1_5/5a,V1_6abc, V1_7abc,V1_9, Mn_1,Mn_2,Mn_3,Mn_4,Mn_5,Mn_6/6a/b/c,Mn_7/7a, Mn_8.Mn_9,Mn_12,Vn_2,Vn_5/5a,Vn_6abc,Vn_7abc, Vn_9,TV1_4_5,TM1_1,TM1_2,TM1_5,TM1_7,TM1_8, TM1_9,TM1_10,TM1_13,TV2_4_5,TM2_1,TM2_2, TM2_5,TM2_7,TM2_8,TM2_9,TM2_10,TM2_12, ALPA_1,ALPA_2,PA_1,PA_2,PA_4,MD_5,HRP_9, MTT2_1,MTT2_2,MTT2_3,MTT2_6,MTT2_7,MTT2_11, EFUrules,ULVT_3,ULVT_4,ULVT_6,ULVT_9, BUMP_4_G,BUMP_5_G,BUMP_7_G,Conventionrules, DUMC_2,DUMC_12,DUMC_18,DUMC_22, 18) Add AA_9b,AA_13a,AA_20,AA_21,AA_22,DNW_8a, DNW_8b,DNW_9,PSUB_14,PSUB_15,DG_15,DG_16, DG_17,DG_18,DG_19,BORDER_6,GT_3d/e,GT_17a, GT_18a,GT_29,GT_30,GT_31,GT_32,GT_33,LVT_15, HVT_15,SP_22,SN_20a,sn_23,SAB_8a/b,CT_13,CT_20, M1_4a,M1_15,M1_16,V1_17a/b,V1_18,V1_19,V1_20, V1_21,Mn_4a,Mn_13,Mn_14,Mn_15,Mn_16,Vn_17a/b, Vn_18,Vn_19,Vn_20,Vn_21,TV1_14,TV1_15,TV1_16, TV1_17,TV2_15,TV2_16,TV2_17,TV2_18,PA_11, MTT2_6a,MTT2_13,ULVT_5,ULVT_10,Inductorrules, LOGO rules,DUMC_40~DUMC_51, 19) Delete GT_20c/d,SAB_15,CT_9,CT_11,M1_11a/b, ALPA_5, ALPA_11, ESD_8, DUMCN rules. Please be kindly noted that the contents of this document may contain the confidential information of a third party such as SMIC's technology licensor, IP vendor, or customer, etc. Pursuant to the agreement between such third party and SMIC, SMIC's employee with a need to access such third party's confidential information shall (i) use such information solely for the benefit of SMIC or the third party, and for no other purpose, (ii) keep such information confidential for a period of fifteen years or longer from its disclosure of such information to SMIC, and (iii) agree that the third party is a third party beneficiary for the confidentiality obligation stated herein.
7	1.11_0	2017-03-01	Penny Hua	1. Update Doc. Rev from 6 to 7 2. Update Tech Dev Rev from 1.10_0 to 1.11_0 3. Update Doc Title to 40nm Generic Logic Low Leakage 0.9V/1.1V/1.2V/1.8V/2.5V Calibre DRC 4. Update item 5 : to TD-LO40-DR-2001 (Doc.Rev: 5 Tech DevRev: 1.11)

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				5. Update item 6 :to Technology R&D / Tech-Bas Des-Enb / Phy. Verification&Yield Enh / DRC 6. Update item 8: DRC version to 2016.2_18.12 7. Update item 9:DRC&ANT name to For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.11_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.ant 8. Update item 11: attachment main rules' DRC runset to SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.tar.gz After decompression: SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.ant 9. For main drc: update GT_32_R/CT_5b follow DR For ant : version up to follow DR version up
8	1.12_0	2017-05-02	Penny Hua	1. Update Doc. Rev from 7 to 8 2. Update Tech Dev Rev from 1.11_0 to 1.12_0 3. Update item1 Doc Title to 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage And Generic Calibre DRC 4. Update item5 reference to LOGIC DESIGN RULE: TD-LO40-DR-2001 (Doc.Rev: 6 Tech DevRev: 1.12) and title 5. Update DRC&ANT name in section 9; For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.ant 6. Update option setting and content in item 9.1 to follow DR update the switch 7. Update item11 attachment main rules' DRC runset to SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.tar.gz After decompression: SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_0

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				.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_0 .drc SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.ant 8. For main drc: update Add LD_12/RESWAA_14/M1_3a/M1_3b/M1_17/M1_18/M1_19/M1_20/Mn_3a/Mn_3b/Mn_17/Mn_18/Mn_19/Mn_20 LU_3d/LU_3g/V1_2a/V1_2b/Vn_2a/Vn_2b/ESDLO_17a/ESDLO_17b/WBPAD_1/WBPAD_2/WBPAD_3/WBPAD_4/WBPAD_5/ WBPAD_6/WBPAD_7/WBPAD_8_R/WBPAD_9WBPAD_10/GRC_13/ Modify GT_31/TM1_7/TM1_10/TM2_7/TM2_10/DGR_3/M1_3/M1_3_R/M1_16/Mn_3/Mn_3_R/Mn_15/Mn_16/MOM_4_G/ ESDLO_7/LU_1/LU_2/MTT2_5/MTT2_7/IND_R_9_G/AA_2/TG_5/TG_11_R/RESAA_12/RESAA_13/DG_5/GT_9/ DNW_3/AA_2/RESNWST_6/CORN_1/BORDER_6/GT_24b_R/GT_24c_R/GT_25b_R/GT_25c_R/GT_33/ GT_34_R_OFF/GT_35_R_OFF/PLRES_6/SP_3a/GRC_1/GRC_3/Convention_1/Convention_2/Convention_3 ESDLO_3a_G/ESDLO_3b_G/all DFM_OFF except EXDFM layer For ANT deck: only version up to follow DR version up
9	1.12_1	2017-08-17	Angela Yu	1. Item 8: Update calibre tool version from 2016.2_18.12 to 2016.3_19.12. 2. Item 9: main rule drc and antenna runset version up: a) Update runset name: For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_1.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_1.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12_1.ant b) Update option setting and content in item 9.1 to follow standard DRC request. 3. Item 11 : Attachment version up : a) Update attachment name to SMIC_CalDRC_40GELGLL_0911121825_V1.12_1.tar.gz

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				b) For main rule drc : 1) Update M1_3a,M1_3b,Mn_3a and Mn_3b(n=2~8). 2) Delete switch for GESD1. 3) Update calibre tool version from 2016.2_18.12 to 2016.3_19.12. 4) Update switch comment per standard request. For antenna rule runset : 1) Only version up.
10	1.12.1_0	2018-08-22	Huan Yang	1.Item 5: update reference document information. 2.Item 6: update org. name to “Design Service/PDK Division/Physical Verification/ DRC&Dummy deck”. 3.Item 9: main rule drc and antenna runset version up: a) Update runset name: For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12.1_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12.1_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.ant 4.Item 11: Attachment version up : a) Update attachment name to SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.tar.gz b) For main rule drc : 1) Update AA_2, AA_4 and LD_11_G. 2) Update resistor definition for LU_?. For antenna rule runset : 1) Only version up.

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1.Title

40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage And Generic Calibre DRC

2.Purpose

This document is intended to be a release note for the users who adopt SMIC 40nm Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage and Generic Calibre DRC runset for Design Rule Checking for their design reference.

3.Scope

This document presents the latest information about Calibre DRC runset for 40nm Low Leakage and Generic process.

4.Nomenclature

DRC runset Design Rule Check runset

5.Reference

This document and the Calibre DRC runset are based on:
LOGIC DESIGN RULE: TD-LO40-DR-2001 (Doc.Rev: 7 Tech DevRev: 1.12.1)
40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V low leakage and generic Design Rule

6.Responsibility:

Design Service/PDK Division/Physical Verification/ DRC&Dummy deck

7. Subject Content:

7.1 FAQ

8. DRC tool versions

Please notice that this runset must be run with calibre 2016.3_19.12.

9. Calibre DRC options set

There are 3 DRC decks in attached package.
For main rules(include ESD and Latch-up rules):
SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_1.drc
SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_1.drc

For antenna rules:

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SMIC_CalDRC_40GELGLL_0911121825_V1.12_1.ant

9.1 Option setting for main rule's DRC deck

```
##################################################
#        Environmental Variables Setting         #
##################################################

#TOP Metal select. The value can be 10,9,8,7,6,5(include double top metals(TM1,TM2/MTT2)).
#Default value is 10.
set TOPMETAL 10                    ;#valid value is: {10,9,8,7,6,5}

VERBATIM {
#DEFINE CONNECT_ALL_RESISTOR 0    //;#valid value is: {0,1}
                                   //;#default 0
                                   //;#The switch (to connect AA resistor, Hi-R resistor, well resistor
between PAD and AA injector) can control the connectivity of resistor.
                                   //;#if 0,the connection is broken;
                                   //;#if 1,the connection is not broken;

#DEFINE DEFINE_PAD_BY_TEXT    1     // ;# valid value is: {0,1}
                                   //;# default is 1
                                   //;# when"?VDD?","?VCC?","?VSS?","?GND?" as text name using
layer ALPATXT on ALPA,
                                   //;# if 0, it is regarded as I/O PAD; if 1, it is regarded POWER PAD.


//*************************************************
//     SMIC internal choice
//*************************************************


//AA resistor layout guideline selection. Default is YES.
//when check AA resistor layout guidelines, please set it to "YES".
//when not to check AA resistor layout guidelines, please set it to "NO".
#DEFINE RESAA_CHECK YES                //valid value is: {YES, NO}


//NW resistor layout guideline selection. Default is YES.
//when check NW resistor layout guidelines, please set it to "YES".
//when not to check NW resistor layout guidelines, please set it to "NO".
#DEFINE RESNW_CHECK YES                //valid value is: {YES, NO}
```

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//Poly resistor layout guideline selection. Default is YES.
//when check Poly resistor layout guidelines, please set it to "YES".
//when not to check Poly resistor layout guidelines, please set it to "NO".
#DEFINE RESP_CHECK YES //valid value is: {YES, NO}

//ESD layout guideline selection. Default is NO.
//when check ESD layout guidelines, please set it to "YES".
//when not to check ESD layout guidelines, please set it to "NO".
#DEFINE ESD_CHECK NO //valid value is: {YES, NO}

//Latch Up Prevention Layout Guidelines selection. Default is YES.
//when check Latch up guidelines, please set it to "YES".
//when not to check Latch up guidelines, please set it to "NO".
#DEFINE LATCH_UP_CHECK YES //valid value is: {YES, NO}

// LDMOS rule selection. Defaults to NO.
// when check LDMOS rules , please set it to "YES";
// when not to check LDMOS rules, please set it to "NO".
#DEFINE LD NO //valid value is: {YES, NO}

//Inductor rule selection. Defaults to YES.
//when check inductor rules , please set it to "YES".
//when not to check inductor rules, please set it to "NO".
#DEFINE INDUCTOR YES //valid value is: {YES, NO}

//BUMP rule selection. Default is "NO".
//when check BUMP rules , please set it to "YES".
//when not to check BUMP rules, please set it to "NO".
#DEFINE BUMP NO //valid value is: {YES, NO}

//LOGO guideline selection. Default is YES.
//when check LOGO guidelines, please set it to "YES".
//when not to check LOGO guidelines, please set it to "NO".
#DEFINE LOGO YES //valid value is: {YES, NO}

//MOM(metal oxide metal) capacitor rule selection for MOM.4[G]. Defaults to YES.
//when check MOM.4[G] guidelines,please set it to "YES";

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//when not to check MOM.4[G] guidelines,please set it to "NO".
#DEFINE MOM_4_CHECK YES //valid value is: {YES, NO}

//MOM(metal oxide metal) capacitor rule selection for MOM.5[G]. Default is YES.
//when check MOM.5[G],guidelines,please set it to "YES".
//when not to check MOM.5[G] guidelines,please set it to "NO".
#DEFINE MOM_5_CHECK YES //valid value is: {YES, NO}

//MOM(metal oxide metal) capacitor rule selection for MOM.6[G]. Default is YES.
//when check MOM.6[G] guidelines,please set it to "YES".
//when not to check MOM.6[G] guidelines,please set it to "NO".
#DEFINE MOM_6_CHECK YES //valid value is: {YES, NO}

//This switch is for default turn on DFM rules.
//Default value is "YES" to check default turn on DFM rules.
//Please change the value "YES" into "NO" if you don't want to check these rules.
#DEFINE DFM_rules_CHECK_ON YES //valid value is: {YES, NO}

//This switch is for default turn off DFM rules.
//Default value is "NO" not to check default turn off DFM rules.
//Please change the value "NO" into "YES" if you want to check these rules.
#DEFINE DFM_rules_CHECK_OFF NO //valid value is: {YES, NO}

//If top metal is layer TM2 (layer 122, top metal with regular thickness), then please use default value NO.
//If top metal is layer MTT2 (layer 231, top metal with ultra thickness), then please change it into YES.
#DEFINE UTM NO //valid value is: {YES, NO}

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//CORN rule selection. Default is YES.
//when check Chip without SEAL RING , please set it to "YES".
//when check Chip with SEAL RING, please set it to "NO".
#DEFINE CHK_CORN YES //valid value is: {YES, NO}

//BORDER rule selection. Default is YES.
//If customer check the chip level , please set it to "YES".
//If customer check the IP level , please set it to "NO".
#DEFINE CHK_BORDER YES //valid value is: {YES, NO}

//SRAM GT direction selection.
//Default is V (vertical).
//But it can also be changed into H (horizontal).
#DEFINE SRAM_GT_DIRECTION V //valid value is: {H,V}

// This switch is only for LU_3_3a_G,LU_3_3b_G,LU_3_3c_G, LU_3_3d_G rules
// default value is "FAST" to fast check LU_3_? rules and reduce runtime,yet it may report false alarm in min space value
// please change the value "FAST" into "ACCURATE",if you want to accurately check LU_3_? rules, yet it have long runtime.
#DEFINE LU_3_CHECK FAST //valid value is: {FAST, ACCURATE}

9.2 Option setting for antenna rule's DRC deck

a.Please set the correct total metal counts in your design.DRC default to check 10 level metal.

```
# Total metal count. Valid choices are 4~10 . See metallization option table in document. Default is 10 .

set M 10
```

For 9 level metal, please set this option to "M 9";
For 8 level metal, please set this option to "M 8";
For 7 level metal, please set this option to "M 7";
For 6 level metal, please set this option to "M 6";
For 5 level metal, please set this option to "M 5";
For 4 level metal, please set this option to "M 4";

b. Please set the correct top metal counts in your design. DRC default to check 2 top metal.

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```
# Top metal count. Valid choices are 1,2. See metallization option table in document. Default is 2 .

set T 2

# If M is 4, then valid choices of T is 1.
# If M is 5, then valid choices of T is 1,2.
# If M is 6, then valid choices of T is 1,2.
# If M is 7, then valid choices of T is 1,2.
# If M is 8, then valid choices of T is 1,2.
# If M is 9, then valid choices of T is 1,2.
# If M is 10, then valid choices of T is 2.
```

For one top metal count, please set this option to “T 1”.

c. Please set the correct top metal in your design. DRC default to check normal top metal,

```
################################
#  <Choice 3>   top metal name  #
################################

# top metal name. Valid choices are TM2, MTT2. TM2 is for gds number 122, MTT2 is for gds number 231. Default is TM2

set TM_NAME TM2
```

For ultra-thickness top metal, please set this option to “TM_NAME TM2”.

d. Please set the correct ALPA thickness in your design. DRC default to check 1.45um thickness ALPA,

```
#################################
#  <Choice 4>   ALPA thickness  #
#################################

# There are two kinds of ALPA thickness for selection: 1.45um and 2.8um, Default to check 1.45um
# If customer want to check 2.8um, please change this option to "set ALPA_thickness 2.8"

set ALPA_thickness 1.45
```

For 2.8um thickness ALPA, please set this option “2.8”

10. Known issues

MTT2_13 may have false alarm in below pattern:

Two TV2 are located on the same TM1 and on the same MTT2, but they are located on the different intersect region of TM1 and MTT2. If the space between the two TV2 is <= 1.7um, it is also a good pattern, but DRC would highlight it.

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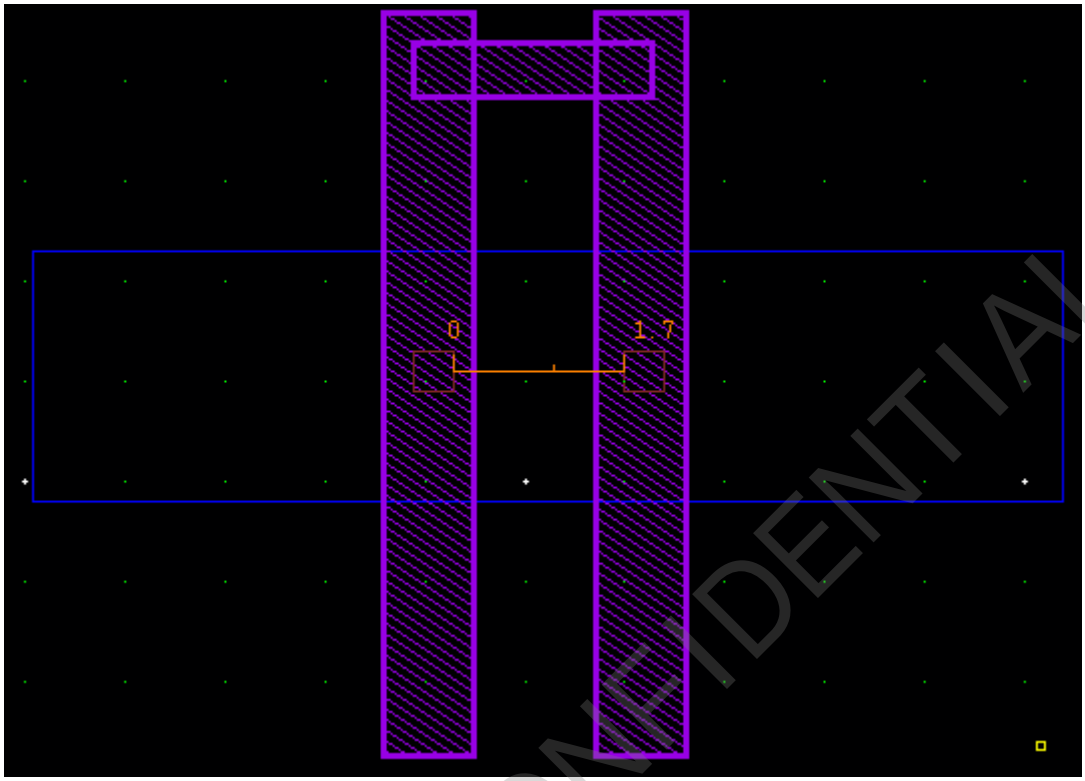

11. Attachment

DRC runset package:
SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.tar.gz

After decompression:
SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12.1_0.drc
SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12.1_0.drc
SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.ant
UserGuide
Release_Note

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Change History:

Document Change History				
Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
0R	1.0	2012-09-28	Liu Li	Initial
1R	1.0	2012-10-29	Liu Li	1. Add “40nm Logic Design Rule check special notes” in item 5 reference. 2. Update attachments: Add BORDER rule switch update input layer exclude MARKS RESAA.14,RESP1.15,NW.12,TG.5,DG.5,GT.6a,SP.20a/21a,CT.3,CT.3a CT.6abd/6acd,CT.7ab,CT.19,M1.3,M1.7e,V1.6abc/7abc,V1.8,V1.17 Vn.17,Mn.12,Vn.6abc/7abc,DFM.5/13/19,DUMCN.21/22,DUP.1/5,MTT2.5
1.1R	1.0	2012-10-30	Liu Li	Add "TD-LO40-DR-2014" in front of "40nm Logic Design Rule check special notes" in main file "5. Reference".
2R	1.0	2013-04-19	Liu Li	1. change from “SmicDR3R_cal40_log_ll_sali_V1.0_1.1R.tar.gz” to “SmicDR3R_cal40_log_ll_sali_V1.0_2R.tar.gz” in item 8.1,due to 40nm DR special notes update. a. In the DRC deck, update RESP1.12,RESP1.13 V1.3,Vn.3,Mn.7d,Mn.7e,V1.17,Vn.17, and modify the SRAM marking layer form DPSRAM to RFSRAM 2. update 40nm_Logic_Calibre_DRC_release_note.pdf in item 8.2, due to item 8.1 update.

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3R	1.0	2013-07-16	Liu Li	1. change from "SmicDR3R_cal40_log_ll_sali_V1.0_2R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_3R.tar.gz" in item 8.1,due to 40nm DR special notes update. a. In the DRC deck, update MTT2_5,SP.3a/3b/4/15,SN.3a/3b/4/15,SRSN.3a/3b 2. update 40nm_Logic_Calibre_DRC_release_note.pdf in item 8.2, due to item 8.1 update. a. In the pdf file, change from "SmicDR3R_cal40_log_ll_sali_V1.0_2R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_3R.tar.gz" in item 1.1, and add MTT2.5 rule description in item 1.6

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4R	1.0	2013-09-18	Liu Li	1. change from "SmicDR3R_cal40_log_ll_sali_V1.0_3R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_4R.tar.gz" in item 8.1,due to 40nm DR special notes update. a. In the *.drc deck, update TG.3/4/5/6/11,ULVT.4,ULVT.6,Mn.6a,Mn.6c,DG.3/4/5/6/11,GT.3c,GT.6,GT.6a, RESAA.5,RESAA.6,AA.4d, DNW.7, PSUB.7,RESP1.5/6/11,SP.14,SN.14,SAB.4/6/7/8/11/15, TM2.9,DUMC.6/16/20/26/26a/28,MTT2.2 Change Mn.12,NW.11,NW.12,AA.C.5 to recommend rule,and set default status in design rule check as off, GT.2b_R,GT.6a_R MD.4_R MD.6_R ESD.9a_R change default status in design rule check as off, Delete ULVT.3,ULVT.5,CT.19,DFM.5,ALPA.11 b. In the *.esd deck, change rules name from ESDLO_5~ESDLO_17 to ESDLO_5_G~ESDLO_17_G 2. update 40nm_Logic_Calibre_DRC_release_note.pdf and 40_LL_Calibre_ESD_release_note.pdf in item 8.2, due to item 8.1 update. a. In the 40nm_Logic_Calibre_DRC_release_note.pdf file, change from "SmicDR3R_cal40_log_ll_sali_V1.0_3R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_4R.tar.gz" in item 1.1, change from "#DEFINE RECOMMENDED YES" to "#DEFINE RECOMMENDED NO"in item 1.2, and add TG.11,ULVT.3~ULVT.6,DFM.5,MTT2.2 rule description in item 1.6 b. In the 40_LL_Calibre_ESD_release_note.pdf file, Change version number from V1.0_1R to V1.0_2R in item 1.5.

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5R	1.0	2013-02-14	Liu Li	1. Change from "SmicDR3R_cal40_log_ll_sali_V1.0_4R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_5R.tar.gz" in item 8.1, due to 40nm DR special notes update. a. In the *.drc deck, Update DNW.2a,DNW.4a coding 2. update 40nm_Logic_Calibre_DRC_release_note.pdf in item 8.2, due to item 8.1 update. In *pdf file, change from "SmicDR3R_cal40_log_ll_sali_V1.0_4R.tar.gz" to "SmicDR3R_cal40_log_ll_sali_V1.0_5R.tar.gz" in item 1.1; update DR/DRC's version number in item 1.5;

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6	1.10_0	2015-09-08	Penny Hua	1. Update Title to 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low leakage and Generic Design Rule Check (DRC) runset of Calibre DRC; 2. Update Purpose description in section 2; 3. Update Reference in section 5; 4. Update responsibility in section 6; 5. Delete DRC tool versions in section 7, for the related content was added into item 8 6. Add DRC tool versions in section 8; 7. Add section 9 “Calibre DRC options set” to explain how to set the switch in DRC runset to customer; 8. Add item 10 for known issues; 9. Update attachment main rules' DRC runset to align with DR, as below: 1) Add switch #DEFINE MOM_5_6_CHECK, #DEFINE MOM_4_CHECK 2) Add rules MOM_4_G,MOM_5_G,MOM_6_G 3) Update below density rules: GT_31_density，M1_15_density，M2_14_density，M3_14_density，M3_16_M1M2M3_density，M4_14_density，M4_16_M2M3M4_density，M5_14_density，M5_16_M3M4M5_density，M6_14_density，M6_16_M4M5M6_density，M7_14_density，M7_16_M5M6M7_density，M8_14_density，M8_16_M6M7M8_density，IND_R_9_G_l_TM2_density,IND_R_9_G_h_TM2_densit, IND_R_9_G_l_MTT2_density, IND_R_9_G_h_MTT2_density 4) Update DNW_8 ALPA_1 LU rules,MTT2_13, BD_6,GT_20b,AA_C_5, Mn_12,Mn_13,GT_33 downgrade to [R] 5) Add DUMC_34a,DUMC_35a,SAB_16_R,NW_1a, TG_16a,GRC_7~GRC_12,LD_10,LD_11 6) Delete ULVT_3,ULVT_5 7) Update DUMC_34,DUMC_35,Mn_2,SRDOP_15, NW_3,ULVT_4,ULVT_6,SAB_5,SAB_6,SAB_8,SAB_11,TG_16,GT_5a,TG_11_R,DUMC_26,CT_1,CT_8, CT_13, DNW_8 8) Move DG_15~DG_19 to TG_16~TG_20

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				9) Update AA_19,GT_20b,SAB_5,Convention_2,DG_16, DG_3,DG_4,TG_3,TG_4,GT_28,GT_32, GT_33, CT_21,SP_8c,SN_7c,SN_8c 10) Delete AA_9b,M1_4a,Mn_4a 11) Update AA_20,DG_18,DG_19,SP_7c,SN_7c, LU_3_3a/b/c/d,AA_15a_R, AA_15b_R, GT_20b, IND_R_8,GT_31, AA_9b, AA_13a, DIFRES_6, DNW_9,NW_11_R,NW_12_R,NW_17_R,RESNWST_6,RESNWAA_9,RESNWAA_11,DG_16,DG_17,DG_18,DG_19, GT_3e,GT_5,GT_5a,SRGT_5,GT_9,SRGT_10,GT_21,GT_28,GT_32,GT_33,PLRES_6,LVT_4,LVT_4a,LVT_5,LVT_5a,HVT_4,HVT_4a,HVT_5,HVT_5a,SP_6,SP_7a/b/c,SN_6,SN_7a/b,CT_5,CT_5_R,CT_20,M1_4a,Mn_4a,V1_19,Vn_19, Vn_23_R,PA_11,ESD_5,MTT2_11,SRAM device rule,LOGO_1,LU_3 12) Delete LVT_15,HVT_15,ULVT_10,SP_22,SN_23, SAB_8a/b, 13) Update switch follow 4R Design rule. 14) Add DFM priority for recommended rules. 15) Update connects follow net definition. 16) Implement ESD and Latch up rules 17) UpdateAA_1,AA_4a,AA_4b_4c,AA_9a,AA_11,AA_13,AA_14,AA_14a,AA_15a,AA_15b,AA_16,AA_17, AA_18,AA_19,RESAA_7,RESAA_12,RESAA_13, AA_C_1,AA_C_4,AA_C_5,DIFRES_6,DIFRES_7, DIFRES_11,DIFRES_12,DNW_1,DNW_2,DNW_2a, DNW_7,NW_1,SRNW_1,NW_3,NW_7,NW_11_R, NW_12_R,SRNW_14,NW_13,NW_14,NW_15, NW_16,RESNWST_6,RESNWST_7,RESNWST_9, RESNWAA_7,RESNWAA_9,RESNWAA_10,RESNWAA_11,PSUB_1,PSUB_4,PSUB_12,TG_1,TG_2,TG_13,TG_14,TG_15,DG_1,DG_2,DG_13,DG_14,BORDER_1,GT_2a,GT_3,GT_3c,SRGT_3,GT_4,SRGT_5,SRGT_6,SRGT_7,GT_8a, GT_17,GT_18,GT_20b,GT_21,GT_27,GT_28,RESP1_7,PLRES_6,PLRES_7,PLRES_8,PLRES_9,PLRES_10,LVT_1,LVT_13,HVT_1,HVT_13,SP_9,SRSP_10,SP_15,SP_20a,SN_7b,SN_9,SN_15,SAB_5,SAB_6,SAB_8,SAB_11,CT_1,CT_3,CT_3a,CT_8,CT_10,M1_1,M1_2,M1_3,SRM1_3sa,M1_4, M1_5,M1_6/6a/b/c,M1_7a,M1_7bc,M1_7e,M1_8/8a,M1_9, M1_10,M1_14,V1_2,V1_5/5a,V1_6abc,V1_7abc,V1_9,

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				Mn_1,Mn_2,Mn_3,Mn_4,Mn_5,Mn_6/6a/b/c,Mn_7/7a, Mn_8.Mn_9,Mn_12,Vn_2,Vn_5/5a,Vn_6abc,Vn_7abc, Vn_9,TV1_4_5,TM1_1,TM1_2,TM1_5,TM1_7,TM1_8, TM1_9,TM1_10,TM1_13,TV2_4_5,TM2_1,TM2_2, TM2_5,TM2_7,TM2_8,TM2_9,TM2_10,TM2_12, ALPA_1,ALPA_2,PA_1,PA_2,PA_4,MD_5,HRP_9, MTT2_1,MTT2_2,MTT2_3,MTT2_6,MTT2_7,MTT2_11, EFUrules,ULVT_3,ULVT_4,ULVT_6,ULVT_9, BUMP_4_G,BUMP_5_G,BUMP_7_G,Conventionrules, DUMC_2,DUMC_12,DUMC_18,DUMC_22, 18) Add AA_9b,AA_13a,AA_20,AA_21,AA_22,DNW_8a, DNW_8b,DNW_9,PSUB_14,PSUB_15,DG_15,DG_16, DG_17,DG_18,DG_19,BORDER_6,GT_3d/e,GT_17a, GT_18a,GT_29,GT_30,GT_31,GT_32,GT_33,LVT_15, HVT_15,SP_22,SN_20a,sn_23,SAB_8a/b,CT_13,CT_20, M1_4a,M1_15,M1_16,V1_17a/b,V1_18,V1_19,V1_20, V1_21,Mn_4a,Mn_13,Mn_14,Mn_15,Mn_16,Vn_17a/b, Vn_18,Vn_19,Vn_20,Vn_21,TV1_14,TV1_15,TV1_16, TV1_17,TV2_15,TV2_16,TV2_17,TV2_18,PA_11, MTT2_6a,MTT2_13,ULVT_5,ULVT_10,Inductorrules, LOGO rules,DUMC_40~DUMC_51, 19) Delete GT_20c/d,SAB_15,CT_9,CT_11,M1_11a/b, ALPA_5, ALPA_11, ESD_8, DUMCN rules.

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7	1.11_0	2017-03-01	Penny_Hua	1.Update Doc. Rev from 6 to 7 2.Update Tech Dev Rev from 1.10_0 to 1.11_0 3.Update Doc Title to 40nm Generic Logic Low Leakage 0.9V/1.1V/1.2V/1.8V/2.5V Calibre DRC 4.Update item 5 : to TD-LO40-DR-2001 (Doc.Rev: 5 Tech DevRev: 1.11) 5.Update item 6 :to Technology R&D / Tech-Bas Des-Enb / Phy. Verification&Yield Enh / DRC 6.Update item 8: DRC version to 2016.2_18.12 7.Update item 9:DRC&ANT name to For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.11_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.ant 8.Update item 11: attachment main rules' DRC runset to SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.tar.gz After decompression: SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.11_0.drc SMIC_CalDRC_40GELGLL_0911121825_V1.11_0.ant 9.For main drc: update GT_32_R/CT_5b follow DR For ant : version up to follow DR version up

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8	1.12_0	2017-05-02	Penny_Hua	1.Update Doc. Rev from 7 to 8 2.Update Tech Dev Rev from 1.11_0 to 1.12_0 3.Update item1 Doc Title to 40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage And Generic Calibre DRC 4.Update item5 reference to LOGIC DESIGN RULE: TD-LO40-DR-2001 (Doc.Rev: 6 Tech DevRev: 1.12) and title 5.Update DRC&ANT name in section 9; For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.ant 6.Update option setting and content in item 9.1 to follow DR update the switch 7.Update item11 attachment main rules' DRC runset to SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.tar.gz After decompression: SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_0.drc SMIC_CalDRC_40GELGLL_0911121825_V1.12_0.ant 8.For main drc: update Add LD_12/RESWAA_14/M1_3a/M1_3b/M1_17/M1_18/M1_19/M1_20/Mn_3a/Mn_3b/Mn_17/Mn_18/Mn_19/Mn_20 LU_3d/LU_3g/V1_2a/V1_2b/Vn_2a/Vn_2b/ESDLO_17a/ESDLO_17b/WBPAD_1/WBPAD_2/WBPAD_3/WBPAD_4/WBPAD_5/ WBPAD_6/WBPAD_7/WBPAD_8_R/WBPAD_9WBPAD_10/GRC_13/ Modify GT_31/TM1_7/TM1_10/TM2_7/TM2_10/DGR_3/M1_3/M1_3_R/M1_16/Mn_3/Mn_3_R/Mn_15/Mn_16/MOM_4_G/ ESDLO_7/LU_1/LU_2/MTT2_5/MTT2_7/IND_R_9_G/AA_2/ TG_5/TG_11_R/RESAA_12/RESAA_13/DG_5/GT_9/ DNW_3/AA_2/RESNWST_6/CORN_1/BORDER_6/GT_24b_R/GT_24c_R/GT_25b_R/GT_25c_R/GT_33/ GT_34_R_OFF/GT_35_R_OFF/PLRES_6/SP_3a/GRC_1/GRC_3/Convention_1/Convention_2/Convention_3 ESDLO_3a_G/ESDLO_3b_G/all DFM_OFF except EXDFM layer For ANT deck: only version up to follow DR version up

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9	1.12_1	Angela Yu	2017-08-17	1. Item 8: Update calibre tool version from 2016.2_18.12 to 2016.3_19.12. 2. Item 9: main rule drc and antenna runset version up: a) Update runset name: For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12_1.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12_1.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12_1.ant b) Update option setting and content in item 9.1 to follow standard DRC request. 3. Item 11 : Attachment version up : a) Update attachment name to SMIC_CalDRC_40GELGLL_0911121825_V1.12_1.tar.gz b) For main rule drc : 1) Update M1_3a,M1_3b,Mn_3a and Mn_3b(n=2~8). 2) Delete switch for GESD1. 3) Update calibre tool version from 2016.2_18.12 to 2016.3_19.12. 4) Update switch comment per standard request. For antenna rule runset : 1) Only version up.

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10	1.12.1_0	Huan Yang	2018-08-22	1. Item 5: update reference document information. 2. Item 6: update org. name to “Design Service/PDK Division/Physical Verification/ DRC&Dummy deck”. 3. Item 9: main rule drc and antenna runset version up: a) Update runset name: For main rules(include ESD and Latch-up rules): SMIC_CalDRC_40GELGLL_0911121825_2TM_V1.12.1_0.drc SMIC_CalDRC_40GELGLL_0911121825_1TM_V1.12.1_0.drc For antenna rules: SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.ant 4. Item 11: Attachment version up : a) Update attachment name to SMIC_CalDRC_40GELGLL_0911121825_V1.12.1_0.tar.gz b) For main rule drc : 1) Update AA_2, AA_4 and LD_11_G. 2) Update resistor definition for LU_?. For antenna rule runset : 1) Only version up.

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